



INCLUDES

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AP[®] Psychology

COURSE AND EXAM DESCRIPTION

Effective
Fall 2024

Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and the AP Psychology Exam and serve as examples of the types of questions that appear on the exam. These sample questions do not represent the full range and distribution of items on an official AP Psychology Exam. After the sample questions is a table that shows which skill, learning objective, and essential knowledge statement each question assesses. The table also provides the answers to the multiple-choice questions.

Section I: Multiple-Choice

1. Oksana experiences pleasurable feelings when she hugs her mother. Recently, her mother started wearing a new perfume, which Oksana can smell when she hugs her mother. When Oksana is shopping, she smells that new perfume near the counter where it is sold. She immediately feels the same pleasurable feelings as she does when she hugs her mother. In terms of classical conditioning, which of the following is the smell of the new perfume?
 - (A) Unconditioned stimulus (UCS)
 - (B) Conditioned stimulus (CS)
 - (C) Positive reinforcement
 - (D) Unconditioned response (UCR)

2. Researchers conducted a study with 200 participants who had been diagnosed with schizophrenia and a comparison group of 200 patients who had not been diagnosed with schizophrenia. The researchers found that participants who had been diagnosed with schizophrenia had significantly larger ventricles¹ than a comparison group. Based on this finding, the researchers concluded that enlarged ventricles cause people to develop schizophrenia. Which of the following most accurately describes why this conclusion is flawed?
- (A) The researchers' sample is not large enough to allow researchers to draw any scientific conclusions.
- (B) The researchers' results indicate no correlation between the variables.
- (C) The researchers' conclusion does not adequately account for the role of GABA in developing schizophrenia.
- (D) The researchers' cause-and-effect conclusions cannot be made because no independent variable is manipulated.

¹Brain cavities filled with fluid.

**Authoritative Parenting Style
Assessment Scores**

7
6
4
9
11
8
9
12
5
9

3. Ten primary caregivers of children completed an assessment to determine the degree to which they practice authoritative parenting. The table shows the participants' scores on this assessment. The lowest possible score is 1, meaning the degree of authoritative parenting is low. The highest possible score is 15, meaning the degree of authoritative parenting is high. Based on the table, what is the range of the caregivers' scores?
- (A) 4
- (B) 8
- (C) 9
- (D) 12

4. Dr. Trenton conducted a study to determine whether massed practice or distributed practice produced better academic outcomes. He recruited volunteers from a high school Spanish class and randomly assigned students to learn a list of 100 new vocabulary words for which they were later given a word recall test. Students prepared for the word recall test using either distributed practice by studying for 30 minutes a day the week before the test, or massed practice by intensively studying the night before the test. What was the dependent variable in this research project?
- (A) Massed practice
 - (B) Distributed practice
 - (C) High school students
 - (D) Word recall

Questions 5 through 7 refer to the following.

Dr. Min conducted a study to investigate creativity. Five volunteers were presented with a series of three objects, one at a time. The volunteers asked to generate as many alternative uses for the objects as possible in five minutes. The data collected on the number of alternative uses each volunteer generated for each object are displayed in the table.

Volunteer	Tweezers	Eraser	Paper Clip
A	0	3	3
B	7	6	5
C	9	12	8
D	4	3	5
E	7	8	11

5. Which of the following would cognitive psychologists use to describe the response of the volunteer who provided the lowest number of alternative uses for the tweezers?
- (A) Achievement
 - (B) Functional fixedness
 - (C) Framing
 - (D) Sunk-cost fallacy
6. What was the median number of alternative uses generated for the eraser?
- (A) 3
 - (B) 5
 - (C) 6
 - (D) 10

7. Which of the following concepts most directly applies to Dr. Min's research question?
- (A) Schema
 - (B) Social loafing
 - (C) Divergent thinking
 - (D) General intelligence (*g*)
8. A doctor is assessing a patient with a sensory disorder. In the first assessment, she asks the patient to touch his nose with his eyes closed. In the next assessment, she asks the patient to walk a short distance with his eyes closed. Which of the following senses is the doctor most likely evaluating?
- (A) Gustation
 - (B) Vision
 - (C) Kinesthesia
 - (D) Touch
9. In the 1960s, a psychologist placed dogs in a box and then exposed them to five trials in which they received an electric shock that they could not avoid. In the sixth trial, the psychologist placed the dogs in a different box. In this trial, the dogs could avoid the shocks by jumping over a small obstacle. None of the dogs jumped over the obstacle. Which of the following concepts explains why the dogs did not jump over the obstacle to escape the shocks in the sixth trial?
- (A) Retroactive interference
 - (B) Learned helplessness
 - (C) Superordinate goals
 - (D) Diathesis-stress
10. Antonella, a native of the Caribbean, learns that a family member died suddenly. She begins to cry uncontrollably, loses control as she yells aggressively, and faints. Which of the following conditions best describes the symptoms Antonella experienced?
- (A) *Ataque de nervios*
 - (B) Agoraphobia
 - (C) *Taijin kyofusho*
 - (D) Dissociative amnesia

11. Which of the following statements best explains why using an experimental research design is inappropriate in studying human development?
- (A) Experiments cannot establish cause-and-effect relationships over time.
 - (B) Direct, long-term manipulation of a child's environment during an experiment is unethical.
 - (C) Experiments with young children are always unethical.
 - (D) Identifying which genetic traits are interacting with the environment is impossible using an experiment.

Questions 12 through 14 refer to the following.

In a study, researchers recruited 10 graduate students at their university to complete a series of tasks in a lab. Participants were asked to rate how desirable they found each item, both a toaster and a microwave, on a scale of 1 to 10, with 1 being not desirable and 10 being very desirable. After completing several complex math problems, participants were told that they could choose either a toaster or a microwave as a prize for participating in the study. After selecting their prize, participants were asked to rate the desirability of both items again. The researchers then compared the participants' ratings of the items at the beginning of the study and the rating that the participants provided after choosing their prize. The results are given in the tables.

Table 1: Participants Who Chose a Toaster as Their Prize

Mean Rating of Toaster		Mean Rating of Microwave	
Before choosing toaster	After choosing toaster	Before choosing toaster	After choosing toaster
3.4	8.8	2.6	1.4

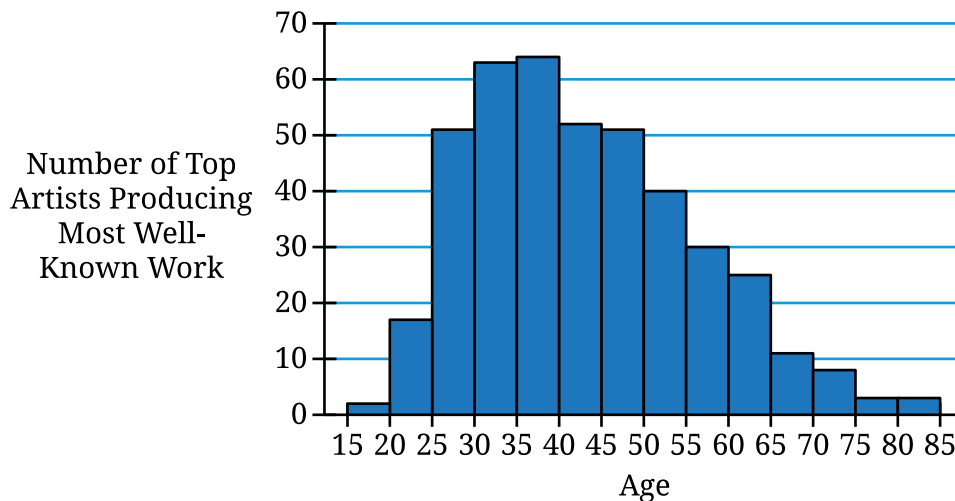
Table 2: Participants Who Chose a Microwave as Their Prize

Mean Rating of Toaster		Mean Rating of Microwave	
Before choosing microwave	After choosing microwave	Before choosing microwave	After choosing microwave
3.4	2	3.8	8.8

12. Which of the following implicit attitudes may have influenced how the participants rated each item?
- (A) A participant being aware that they need one item more than the other
 - (B) A participant seeing the cost of each item and knowing that one is more expensive than the other
 - (C) A participant attributing a greater value to one item over the other without knowing why
 - (D) A participant seeing that the toaster is their favorite color

13. Which of the following changes to the research design will expand the generalizability of the study's findings to a larger population?
- (A) Including participants who are not graduate students
 - (B) Using a correlational design to determine causality
 - (C) Including a debriefing session after the study to inform participants
 - (D) Increasing the prize value so that the participants are more motivated
14. Which of the following concepts is demonstrated by the results of the study?
- (A) The testing effect
 - (B) Social desirability
 - (C) Cognitive dissonance
 - (D) Fundamental attribution error
15. When 90-year-old Mrs. Glass socializes at her senior activity center, she gravitates toward activities that she can still perform well, which also demonstrates that her crystallized intelligence is functioning. Which of the following activities will most likely appeal to Mrs. Glass?
- (A) A "name that song" game featuring songs from her childhood
 - (B) A trivia game featuring current movies
 - (C) A new video game in which aliens fly a spaceship
 - (D) A class teaching her how to use the latest computer technology

16. Dr. Faley conducted a study of 100 patients diagnosed with adult attention-deficit/hyperactivity disorder (ADHD). The first 50 patients identified for the study were given a new ADHD medication, while the rest were given a placebo. Patients received EEG scans before and after the treatment. The scans showed more typical brain activity in the participants who were given the medication than the participants who were given the placebo. What should Dr. Faley have done to avoid potential confounding variables in the study?
- (A) Operationally defined the meaning of adult ADHD
 - (B) Chosen people who had not been diagnosed with adult ADHD to take the medication as well
 - (C) Randomly assigned which patients got the medication and which patients got the placebo
 - (D) Using qualitative measures to collect data



17. The data in the graph display the ages at which top artists, including painters, writers, and composers, produce what is generally agreed upon to be their most well-known work. Based on the data in the graph, which of the following statements is true?
- (A) More than half of the artists do their most well-known work before age 30.
 - (B) There is a positive correlation between artists' ages and the production of their most well-known work.
 - (C) The number of top artists producing their well-known work peaks at an age in the late thirties and then gradually declines.
 - (D) Artists are more likely to generate their most well-known work after the age of 65.

18. A researcher noticed that Ryan, a young child, learned to write his name better when he was given the opportunity to respond to people around him. Ryan appeared to learn the most when more experienced people assisted him. Which of the following concepts best describes this pattern of development?
- (A) Zone of proximal development
 - (B) Egocentrism
 - (C) Autonomy
 - (D) Object permanence

Questions 19 through 21 refer to the following.

Researchers investigated the extent to which 30 pairs of identical and fraternal female twins shared with their twin a diagnosis for the eating disorder anorexia nervosa. Identical twins are genetically identical, and share the same DNA pattern. Fraternal twins are only as genetically similar as any sibling pair. Researchers found that 9 out of 16 pairs of the identical twins were diagnosed with anorexia nervosa, while 1 out of 14 pairs of the fraternal twins were diagnosed with anorexia nervosa.

Source for research study:

Holland, A.J., Hall, A., Murray, R., Russell, G.F., & Crisp, A.H. (1984). Anorexia nervosa: a study of 34 twin pairs and one set of triplets. *British Journal of Psychiatry*, 145, 414-419.

19. Which of the following explains why researchers could not implement an experimental design in this study?
- (A) The twins could not be given a placebo.
 - (B) The twins could not be randomly assigned to groups.
 - (C) The twins could not be studied in an in-depth way individually.
 - (D) Multiple studies of the twins could not be conducted over a long period and combined.
20. Which of the following conclusions can be made based on the data collected in this study?
- (A) Identical twins both developing anorexia nervosa is no more likely than fraternal twins.
 - (B) Heredity appears to be correlated with the development of anorexia nervosa.
 - (C) One twin developing anorexia nervosa will cause the other twin to develop the disorder.
 - (D) Environment appears to be more strongly correlated with the development of anorexia nervosa than heredity.

21. Which of the following psychological perspectives would best help researchers answer questions about the relationship between heredity and the environment regarding the development of disorders such as anorexia nervosa?
- (A) Cognitive
 - (B) Psychodynamic
 - (C) Humanistic
 - (D) Biological

Section II: Free-Response

The following are examples of the free-response question types found on the exam. Scoring rubrics for each question follow the sample question.

QUESTION 1: ARTICLE ANALYSIS QUESTION (AAQ)

Your response to the question should be provided in six parts: A, B, C, D, E, and F. Write the response to each part of the question in complete sentences. Use appropriate psychological terminology in your response.

Using the source provided, respond to all parts of the question.

- (A) Identify the research method used in the study.
- (B) State the operational definition of executive functioning.
- (C) Describe what the difference in means indicates for the immediate recall task for the multivitamin group as compared to the placebo group.
- (D) Identify at least one ethical guideline applied by the researchers.
- (E) Explain the extent to which the research findings may or may not be generalizable using specific and relevant evidence from the study.
- (F) Explain how at least one of the research findings supports or refutes the researchers' hypothesis that taking a multivitamin slows cognitive decline in later life.

Source

Introduction		
Few large-scale, long-term studies have been conducted to test whether taking a multivitamin makes a difference in improving memory ability as one ages. In this study, researchers examined whether taking a multivitamin slows cognitive decline in later life.		
Participants		
An earlier study, which was conducted in 2017 and included over 21,000 people, examined the effects of taking a multivitamin on health outcomes. From that study's sample, over 7,000 people received a mailed invitation to participate in this study. Of those who received the invitation, almost 4,000 participants agreed to participate and were accepted. To be accepted, participants had to be over 65 year of age if women and over 60 years of age if men. In addition, they could not participate if they had ever had a stroke, if they had received a cancer diagnosis in the two years before the study, or if they had a history of any other serious illnesses. Participants had to be able to communicate in English and have access to an Internet-connected computer. A computer randomly assigned participants to two groups. Participants in Group 1 received a pack of multivitamins each month by mail to take one pill twice a day. Participants in Group 2 received a pack of placebo pills in the same type of packaging as Group 1 and with the same instructions. The sample size of Group 1 was 1,758 people, and the sample size of Group 2 was 1,804 people. The demographics of each group are listed in the table:		
Demographic	Group 1 – Multivitamin	Group 2 – Placebo
	Percent or Mean (Standard Deviation in parenthesis when reported)	Percent or Mean (Standard Deviation in parenthesis when reported)
Age	70.9 (4.5)	71.0 (4.6)
Gender		
Men	32.9%	33.4%
Women	67.1%	66.6%
Race/Ethnicity		
White	93.1%	93.5%
African American	2.3%	2.6%
Hispanic	1.5%	1.4%
Other race or Multiple races	1.8%	1.2%
Asian or Native Hawaiian	0.2%	0.1%
American Indian or Alaska Native	1.1%	1.2%

Method															
Once a year for 3 years, participants were asked to complete an online test to evaluate episodic memory and executive functioning. Instructions on how to access the test materials were emailed to participants, and participants who responded to the email indicated their consent. The participants received a \$15 gift card for each annual assessment, regardless of completion. The multivitamin used is widely available in the United States. Side effects of taking the multivitamin include low rates of stomach pain, diarrhea, skin rash, bruising, and an increased rate of gastrointestinal bleeding, which are considered normal side effects for those taking a multivitamin in the general population. Two different tasks were used to measure episodic memory and executive functioning in this study: ▪ To measure episodic memory, participants completed a recall task in which they were first shown a set of words, presented one at a time for three seconds each. They were then asked to recall the set of words, once immediately after the word list was presented and again after 15 minutes had passed. Participants who recalled more words correctly earned higher scores on this test. ▪ To measure executive functioning, participants were first shown one set of items and were then shown a second set of items. They were asked to identify whether items in the second set of items were the same as or different from the first set of items they were shown. Participants earned higher scores the more quickly they correctly identified whether the items from the second set of items were the same as or different from those in the original set.															
Results and Discussion															
Compared with participants taking a placebo, participants receiving the multivitamin had significantly greater improvement in the recall task at the end of the first year. Performance on the immediate recall memory task in Group 1 improved from a mean of 7.10 words at baseline to 7.81 words after 1 year of taking the multivitamin, an improvement mean of 0.71. In Group 2, which received the placebo pills, performance on the immediate recall memory task improved from a mean of 7.21 words at baseline to 7.65 words after 1 year, an improvement mean of 0.44. When comparing the multivitamin group with the placebo group, averaged across all 3 years of intervention, findings suggest that the memory improvement is sustained over time, as shown in the graph. Comparison of Memory Improvement Between 2 Groups <table><thead><tr><th>Time</th><th>Multivitamin</th><th>Placebo</th></tr></thead><tbody><tr><td>Baseline</td><td>0.0</td><td>0.0</td></tr><tr><td>Year 1</td><td>0.71</td><td>0.44</td></tr><tr><td>Year 2</td><td>0.95</td><td>0.75</td></tr><tr><td>Year 3</td><td>1.15</td><td>0.90</td></tr></tbody></table> Researchers estimate that the effect of the multivitamin intervention improved memory performance in participants in the multivitamin group by the equivalent of 3.1 years of age-related memory change as compared to participants in the placebo group. Researchers also found that executive functioning was not significantly impacted by taking a multivitamin. The findings suggest that the greatest benefit to taking a multivitamin is found in immediate memory recall, something especially vulnerable in aging adults.	Time	Multivitamin	Placebo	Baseline	0.0	0.0	Year 1	0.71	0.44	Year 2	0.95	0.75	Year 3	1.15	0.90
Time	Multivitamin	Placebo													
Baseline	0.0	0.0													
Year 1	0.71	0.44													
Year 2	0.95	0.75													
Year 3	1.15	0.90													

Adapted from *The American Journal of Clinical Nutrition*.

Reprinted from *The American Journal of Clinical Nutrition*, 118, Yeung, L-K., Alschuler, D. M., Wall, M., Luttman-Gibson, H., Copeland, T., Hale, C., Sloan, R. P., Sesso, H. D., Manson, J. E., Brickman, A. M. Multivitamin supplementation improves memory in older adults: A randomized clinical trial, pages 273–282, Copyright 2023, with permission from Elsevier.

QUESTION 2: EVIDENCE-BASED QUESTION (EBQ)

This question has three parts: Part A, Part B, and Part C. Use the three sources provided to answer all parts of the question.

For Part B and Part C, you must cite the source that you used to answer the question. You can do this in two different ways:

- Parenthetical Citation:

For example: “...(Source 1)”

- Embedded Citation:

For example: “According to Source 1,…”

Write the response to each part of the question in complete sentences. Use appropriate psychological terminology.

Using the sources provided, develop and justify an argument about the best time for school to start for students in grades 6 to 12.

- Propose a specific and defensible claim based in psychological science that responds to the question.
- Support your claim using at least one piece of specific and relevant evidence from one of the sources.
 - Explain how the evidence from Part B (i) supports your claim using a psychological perspective, theory, concept, or research finding learned in AP Psychology.
- Support your claim using an additional piece of specific and relevant evidence from a different source than the one that was used in Part B (i).
 - Explain how the evidence from Part C (i) supports your claim using a different psychological perspective, theory, concept, or research finding learned in AP Psychology than the one that was used in Part B (ii).

Source 1

Introduction
Researchers investigated the effects of changes in school start times on students in elementary, middle, and high school. They were particularly interested in how school start times related to factors such as academic performance and sleep.
Participants
Eight school districts in Minnesota, and a total of 38,019 students, were involved in this study. Most of the students were white (54.7%), with 12.4% Hispanic, 8.8% Black, 8.6% Asian, 4.9% multiracial, 3.6% Somali, 3.1% American Indian, and 2.1% Hmong ² . The sample consisted of 49.2% women and 50.8% men. The grade levels of students were evenly divided (i.e., 25.6% grade 5, 27.3% grade 8, 26.1% grade 9, and 20.9% grade 11).
Method
Four of the school districts observed shifted to a later school start time, and the four other districts kept an earlier school start time. Two variables were measured in this study. The first was the grade point average (GPA) of the students. Students reported the course grades they generally received during their current school year (e.g., Mostly As, Mostly Bs, etc.). The responses were converted to values on a 4.0 scale (mean = 3.2, standard deviation = 0.9). The second measured variable was the amount of sleep students generally got on a school night and whether they met the recommended hours of sleep for their grade level.
Results and Discussion
Results showed that students in school districts that moved to a later start time reported higher GPAs and that higher proportions of those students reported obtaining the recommended hours of sleep as compared with students in districts with the earlier school start time. The increase in GPA for each grade in the districts with a later school start time (that is, comparing grade 5, grade 8, grade 9, and grade 11) over the districts with the earlier school start time ranged from 0.07 to 0.13 GPA points. The probability of meeting the recommended hours of sleep increased by 16% for students in districts with a later school start time, a statistically significant finding. However, for students in districts with a later school start time, the likelihood of students obtaining the recommended hours of sleep increased by 35% in grade 5, 11% in grade 8, 8% in grade 9, and 5% in grade 11. Only the grade 8 increase was statistically significant. The likelihood of obtaining the recommended hours of sleep for students in districts with an early school start time did not change at statistically significant levels.

Adapted from *American Educational Research Association*.

Caesar, J., Lamm, R., Rodriguez, M.C., & Heistad, D.J. (2021, April 10). Changes in school start time have a significant effect in the amount of sleep and reported grade point average of students [Paper presentation]. *American Educational Research Association Annual Meeting*. <https://conservancy.umn.edu/handle/11299/194887>

²an ethnic group that lives primarily in Southwest China and in countries in Southeast Asia such as Thailand, Laos, Myanmar, and Vietnam

Source 2

Introduction
Most adolescents in the United States do not obtain sufficient sleep. Early school start times play a significant role in adolescent sleep deprivation. Most primary and secondary schools begin classes earlier than 8:30 am. Perceived barriers to implementing a delayed school start time have been suggested in the literature but have not been measured. This study explored both the potential barriers to implementing later high school start times, and other, facilitating factors that may encourage adoption of this practice.
Participants
A convenience sample of 116 respondents participated in the survey, with a response rate of 7.2%. The participants' job titles included superintendents (7.8%; 9 participants), assistant superintendents (16.4%; 19 participants), principals (20.7%; 24 participants), assistant principals (42.2%; 40 participants), counselors (11.2%, 13 participants), and school board members (1.7%; 2 participants). At the time of this study (2017), all worked in a high school that was identified as having implemented a delayed school start time.
Method
School administrators who had delayed their school start times were invited to complete an online questionnaire ranking the perceived barriers and facilitating factors for implementing the delayed start times. Participants were asked if they considered each barrier question as "significant," "moderate," "minor barrier," or "not a barrier" to interfere with implementing a delayed school start time. These statements referred to transportation issues, extracurricular considerations, stakeholder groups who were resistant to change, and additional factors such as time for homework and amount of time waiting for the bus. Participants were also asked if each facilitating factor statement was "significant," "moderate," "minor," or "not a facilitating factor" to support implementing a delayed school start time. These statements addressed stakeholder involvement and education, scheduling alternatives, transportation, and other factors, such as spending less unsupervised time at home. The questionnaire contained the following open-response questions: • "Do you offer classes that students can elect to take before the delayed school start time (sometimes called "zero-hour classes)?" • "Did your district 'flip' elementary and high school start times?" • "Would you recommend that other districts pursue delaying a school start time?"

Results and Discussion

Each statement received a “Barrier Percentage” or “Facilitating Factor Percentage” score, which was the percent of adults in each category that perceived the statement as a significant, moderate, or minor factor in discouraging or encouraging implementing the delayed start times.

Among the most cited perceived barriers were a tiered bus system (in which one bus serves several schools), school athletes missing more afternoon classes, and less time after school for athletics. The most cited facilitating factors were school administrator involvement in the decision-making process and sleep education for family members and school administrators.

Results from this analysis are in the tables:

Distribution of Top 10 Perceived Barriers for School Start Time Delay	
Barrier	Barrier Percentage
Athletes missing more afternoon classes to attend or travel to games	84.2
Less time after school for athletic activities	80.7
Family members resistant to change schedule	78.9
Unavailability of adolescents to provide after-school care for siblings	75.6
Teachers resistant to change schedule	74.8
Use of a tiered school bus transportation system	71.1
Elementary students having to wait for bus pickup in the early morning	69.6
Less time for adolescents to work	65.8
Student resistant to changing their schedule	63.5
Increased school transportation costs	58.4

Distribution of Top 10 Facilitating Factors for School Start Time Change	
Facilitating Factor	Facilitating Factor Percentage
Involvement of the teachers in the decision-making process	84.5
Involvement of the school administrators in the decision-making process	84.4
Providing education on adolescent sleep patterns—for students	83.5
Providing education on adolescent sleep patterns—for family members	82.5
Providing education on adolescent sleep patterns—for teachers	81.5
Involvement of the family members in the decision-making process	80.9
Involvement of students in the decision-making process	79.1
Providing education on adolescent sleep patterns—for school administrators	77.1
Ability to schedule elementary start times to maintain bus tier system	74.5
Involvement of support staff in the decision-making process	70.4

Participants found that providing sleep education to fellow administrators, teachers, school staff members, families, and students, and including them in the decision-making process positively facilitated the implementation of delayed school start times. Perceived barriers to implementation may be overcome with support from stakeholders and planning committees.

When the 116 participants were asked if they would recommend a delayed school start time, 69% (80 participants) said they would recommend that other districts pursue a delayed start time; 19.8% (23 participants) said they would not recommend a delayed start time; and 11.2% (13 participants) provided no answer.

Adapted from *Journal of School Health*.

Fitzpatrick, J.M., Silva, G.E., Vana, K.D. (2020). Perceived barriers and facilitating factors in implementing delayed school start times to improve adolescent sleep patterns. *Journal of School Health*, 91(2). 94-101. <https://doi.org/10.1111/josh.12983>.

Source 3

Introduction
The American Academy of Pediatrics developed the following statement on school start times. A committee of scientists and practitioners who specialize in pediatric health issues intended the statement to offer advice on best practices for school start times.
Statement
The American Academy of Pediatrics recognizes that insufficient sleep is an important issue for adolescents. While several factors may influence students' ability to get enough sleep, evidence strongly suggests that earlier school start times contribute to sleep deprivation. The American Academy of Pediatrics strongly supports efforts to set school start times for middle and high school students in a way that helps them get 8.5 to 9.5 hours of sleep a night to improve physical and mental health, safety, academic performance, and quality of life. Biologically, most adolescents experience a "phase delay" in their sleep-wake cycle when they start going through puberty. As compared to their middle childhood sleep-wake cycle, adolescents experience a shift of up to 2 hours in their sleep-wake cycle, which means they fall asleep later. Insufficient sleep hurts academic performance. One study found that 28% of students reported falling asleep in school at least once a week, and more than 1 in 5 students fell asleep doing homework at least once a week. Many studies show an association between decreased sleep duration and lower academic achievement, as well as higher rates of absenteeism and tardiness and decreased readiness to learn. An increase in anxiety and mood disorders in adolescents has also been linked to poor quality and deprivation of sleep. Other specific health-related effects of sleep loss include increased use of stimulants (e.g., caffeine, prescription medications) to counter the effects of chronic sleepiness on academic performance. Adolescents are also at greater risk of drowsy driving-related crashes because of sleep deprivation. According to the US Department of Education statistics for 2011–2012, approximately 43% of the over 18,000 public high schools in the United States currently have a start time before 8:00 AM. The ongoing debate among school districts in the United States regarding later start times for middle and high schools continues to spark controversy. Perceived barriers to changing school schedules include shorter time for athletic practices and interference with scheduling of games, reduced after-school employment hours for students, challenges in providing child care for younger siblings, adjustments in parent and family schedules, potential safety issues, effects on sleep duration in younger children if elementary school schedules are "flipped" with those of middle/high school students, and the need to make alternative transportation arrangements.

Adapted from American Academy of *Pediatrics*.

Au, R., Carskadon, M., Millman, R., Wolfson, A., Braverman, P.K., Adelman, W.P., Breuner, C.C., Levine, D.A., Marcell, A.V., Murray, P.J., O'Brien, R.F., Devore, C.D., Allison, M., Ancona, R., Barnett, S.E., Gunther, R., Holmes, B., Lamont, J.H., Minier, M., Okamoto, J.K., Wheeler, L.S.M., & Young, T. (Adolescent Sleep Working Group, Committee on Adolescence, Council on School Health) (2014). School start times for adolescents. *Pediatrics*, 134(3), 642–649. <https://doi.org/10.1542/peds.2014-1697>

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Essential Knowledge
1	B	1.A	3.7.A	3.7.A.2
2	D	2.C	5.4.B*	5.4.B.2*
3	B	3.B	3.6.A*	3.6.A.2*
4	D	2.B	2.4.A*	2.4.A.4*
5	B	3.A	2.2.A	2.2.A.8
6	C	3.B	2.2.A*	2.2.A.8*
7	C	1.A	2.2.A	2.2.A.8
8	C	1.A	1.6.G	1.6.G.2
9	B	1.A	3.8.A	3.8.A.4
10	A	1.B	5.4.E	5.4.E.1
11	B	2.D	3.1.B	N/A
12	C	1.A	4.2.A	4.2.A.1
13	A	2.B	4.2.A*	4.2.A.1*
14	C	3.A	4.2.B	4.2.B.2
15	A	1.A	3.4.A	3.4.A.3
16	C	2.B	5.5.A*	5.5.A.1*
17	C	3.C	2.2.A*	2.2.A.8*
18	A	1.A	3.4.A	3.4.A.2
19	B	2.A	1.1.A	1.1.A.3
20	B	2.C	1.1.A	1.1.A.3
21	D	1.A	1.1.A	1.1.A.3
Free-Response Question	Question Type		Practices	Primary Unit
1	Article Analysis Question (AAQ)		2, 3, 4	2
2	Evidence-Based Question (EBQ)		1, 4	1

Further exam resources can be found on the AP Psychology Exam Page on AP Central.

*Skills-focused question. The Learning Objective and Essential Knowledge correspond to the stimulus.

Scoring Rubric for Article Analysis Question (AAQ)

FRQ 1: Article Analysis Question (AAQ)			7 Points
Reporting Category	Scoring Criteria		
Part A Research Method (0-1 points)	0 points Does not accurately identify the research method used in the study	1 point Accurately identifies the research method used in the study	
	Decision Rules and Scoring Notes		
	Responses that earn 0 points: - The response does not accurately identify the research method used in the study. - The response includes no identification of the research method used in the study.	Responses that earn 1 point: The response accurately identifies the research method used in the study as an experiment.	
	Examples that earn 0 points: "The research is a correlational study." "The researchers conducted a case study."	Examples that earn 1 point: "The research method is an experiment." "The researchers used an experiment to study the impact of taking a multivitamin on memory recall."	
Reporting Category	Scoring Criteria		
Part B Research Variable (0-1 points)	0 points Does not state a measurable or quantifiable definition of the identified variable used in the study	1 point States a measurable or quantifiable definition of the identified variable as used in the study	
	Decision Rules and Scoring Notes		
	Responses that earn 0 points: - The response states a definition of the identified variable in the study that is not measurable or quantifiable. - The response does not state a measurable or quantifiable definition of the identified variable as used in the study.	Responses that earn 1 point: - The response states that the operational definition of executive functioning is whether the participant correctly identified the second set of objects shown as being the same or different more quickly. - The response states that the operational definition of executive functioning is the scores on the test.	
	Examples that earn 0 points: "Executive functioning is not helped by multivitamins." "Executive functioning is how many words they recall."	Examples that earn 1 point: "The operational definition of 'executive functioning' is if the person was able to identify correctly the second set of objects shown as being different or the same quickly." "The operational definition is the scores on the executive functioning test."	

Reporting Category	Scoring Criteria	
Part C Statistic Interpretation (0-1 points)	0 points Does not accurately describe the identified statistic in relation to the study	1 point Accurately describes the identified statistic in relation to the study
	Decision Rules and Scoring Notes	
	Responses that earn 0 points: - The response restates the differences in the means for the immediate recall task between the multivitamin group and the placebo group without directly relating it to the study. - The response does not accurately describe what is indicated by the differences in the means for the immediate recall task between the multivitamin group and the placebo group in relation to the study. - The response provides a definition of the differences in the means for the immediate recall task between the multivitamin group and the placebo group but does not accurately describe its meaning in relation to the study.	Responses that earn 1 point: - The response accurately describes what the difference in means indicates for the immediate recall task for the multivitamin group as compared to the placebo group in relation to the study. - The response accurately describes that the means show that the multivitamin group recalled more than the placebo group.
	Examples that earn 0 points: <i>"The difference in the means says that they are different."</i> [The response does not describe what the difference in means indicates.] <i>"The mean for the multivitamin group in Year 3 is 1.1, and the mean for the placebo group was 0.9."</i> [The response does not describe what the difference in means indicates.] <i>"The mean is the average of a group of data."</i>	Examples that earn 1 point: <i>"The means show that the multivitamin group had better recall than the placebo group."</i> <i>"The means show that the placebo group had worse recall than the multivitamin group."</i> <i>"The mean for the multivitamin group in Year 3 is 1.1, and the mean for the placebo group was 0.9. This means that the multivitamin group remembered more over time than the placebo group."</i>
Reporting Category	Scoring Criteria	
Part D Ethical Guidelines (0-1 points)	0 points Does not accurately identify at least one ethical guideline applied by researchers in the study	1 point Accurately identifies at least one ethical guideline applied by researchers in the study
	Decision Rules and Scoring Notes	
	Responses that earn 0 points: - The response does not identify an ethical guideline applied by researchers in the study. - The response identifies an ethical guideline that researchers should have applied but was not mentioned in the study. - The response identifies an ethical guideline not applied by the researchers in the study. - The response identifies other features of the study that are not ethical guidelines.	Responses that earn 1 point: - The response identifies that the researchers used informed consent. - The response identifies that the researchers considered that the side effects of taking multivitamins did not constitute undue harm to the participants.
	Examples that earn 0 points: <i>"The researchers debriefed the participants."</i> <i>"The researchers gave participants a \$15 gift card."</i>	Examples that earn 1 point: <i>"The researcher used informed consent with his participants."</i> <i>"Even though there are side effects for taking multivitamins, they were considered normal for most people and not harmful."</i>

Reporting Category	Scoring Criteria	
Part E Generalizability (0-1 points)	0 points Does not propose a claim regarding the generalizability of the study to a population (general or specific). OR Does not use specific and relevant evidence that references participant variables which would impact the generalizability of the study.	1 point Explains the extent to which the study is generalizable using specific and relevant evidence referencing participant variables from the study
	Decision Rules and Scoring Notes	
	Responses that earn 0 points: - The response proposes no claim regarding the generalizability of the conclusion of the study. - The response proposes a claim regarding the generalizability of the conclusion of the study but makes no reference to a population (general or specific). - The response uses evidence that does not apply to the generalizability of the conclusion of the study (e.g., reliability, validity, sample size).	Responses that earn 1 point: - The response proposes a claim about generalizability that references a population (the "larger population" or a population relevant to the study's participants) to explain the extent of the generalizability of the study. - The response proposes that the study is or is not generalizable to the population relevant to the study because of specific and relevant evidence derived from the participant information from the study.
	Examples that earn 0 points: "The study is generalizable." "The study is generalizable to the larger population." "The study used both men and women." "The study is generalizable because everyone takes vitamins."	Examples that earn 1 point: "The study is generalizable to both men and women because both were used in the study." "The study is generalizable to older adults only because that is the only group the researchers used."
	Additional Note: If the response asserts the study is "partly" generalizable or asserts the study is both generalizable and not generalizable, their evidence must support their assertion by showing ways the study is both generalizable and not generalizable to earn 1 point.	

Reporting Category	Scoring criteria		
Part F Argumentation (0-2 points)	0 points Does not accurately explain how the results of the study support or refute the psychological concept or hypothesis presented in the question.	1 point Uses the results of the study but does not explain how the psychological concept or hypothesis is supported or refuted OR Explains that the psychological concept or hypothesis is supported or refuted but does not use any results from the study	2 points Uses a specific result from the study to explain how the results support or refute the psychological concept or hypothesis presented in the question. The results are accurately interpreted.
	Decision Rules and Scoring Notes		
	Responses that earn 0 points: - The response does not explain how at least one of the research findings support or refute that taking a multivitamin slows cognitive decline in later in life. - The response does not accurately explain how at least one of the research findings support or refute that taking a multivitamin slows cognitive decline in later in life.	Responses that earn 1 point: - The response proposes that the evidence alone is sufficient to demonstrate that taking a multivitamin slows cognitive decline in later in life without additional explanation. - The response uses at least one of the research findings to explain how taking a multivitamin slows cognitive decline later in life. However, the interpretation of the results of the study includes some inaccuracies.	Responses that earn 2 points: The response uses at least one of the research findings to accurately explain how taking a multivitamin slows cognitive decline later in life. The results of the study are accurately interpreted.
	Examples that earn 0 points "The study shows that executive functioning is not affected by taking a multivitamin, so it's not worth it to take them." "The study shows that people should take a multivitamin." "Older adults should not take a multivitamin because of the side effects."	Examples that earn 1 point: "Older adults should take a multivitamin because it improves their memory and thinking." [The response provides an explanation without using results.] "Older adults gain 3.1 years of memory change because of taking a multivitamin." [The response uses results from the study without further explanation.] "Older adults should take a multivitamin because it improves their memory and executive functioning." [The response uses evidence but includes an inaccurate interpretation.]	Examples that earn 2 points: "This study shows that older adults can delay some cognitive decline if they take a multivitamin because it seems to improve episodic memory, but not executive functioning." "The study shows that taking a multivitamin does not totally prevent cognitive decline in old age because executive functioning is not significantly improved by taking one."

Scoring Rubric for Evidence-Based Question (EBQ)

FRQ 2: Evidence-Based Question (EBQ)		7 Points
Reporting Category	Scoring Criteria	
Part A Claim (0-1 points)	0 points Does not propose a claim that is relevant to the question	1 point Proposes a claim that is relevant to the question
	Decision Rules and Scoring Notes	
	Responses that earn 0 points: - The response describes the question without making a claim. - The response proposes a claim unrelated to the question. - The response proposes an oversimplified conclusion or evidence from a provided source.	Responses that earn 1 point: - The response proposes a claim that suggests a positive or negative effect on the question. - The response proposes a recommendation for the application of the question. - The response proposes a specific effect on the question.
	Examples that earn 0 points: "Determining a school start time for secondary students is difficult." "Sleep is important for teenagers." "Students should get more sleep."	Examples that earn 1 point: "School for secondary students should start later in the day." "School for secondary students should start some time before 8:00 a.m." "Secondary school students should start school at 9 a.m." "Secondary school students should start school two hours later than elementary students."
	Additional Notes: A claim that meets the criteria can be awarded the point regardless of whether the responses in Parts B and C successfully support the claim.	

Reporting Category	Scoring Criteria	
Part B (i) Evidence (0-1 points)	0 point Does not identify nor correctly cite one piece of specific evidence from one of the provided sources to support the claim. Any evidence provided is not relevant to the question.	1 point Uses one piece of correctly cited, specific, and relevant evidence from one of the provided sources to support the claim
	Decision Rules and Scoring Notes	
	Responses that earn 0 points: - The response identifies no evidence. - The evidence is not correctly cited. - The evidence is nonspecific. - The response provides evidence not relevant to the claim.	Responses that earn 1 point: The evidence is correctly cited and provides specific evidence relevant to the claim.
	Examples that earn 0 points: "Sources say secondary school start time should be 9:00 a.m." [no citation] "Secondary school start time should be 9 a.m. because people say taking tests before 9 a.m. is very difficult for teenagers." [The response has not provided a citation.]	Examples that earn 1 point: "Secondary school start time should be 9:00 a.m. 28% of students reported falling asleep in school at least once a week (Source 1)."
	Additional Notes: Responses that use a correct citation style other than the methods prescribed by the question can earn this point for citing the source.	

Reporting Category	Scoring Criteria		
Part B (ii) Explanation and Application (0-2 points)	0 points Does not explain the relationship between the evidence and the claim	1 point Explains the relationship between the evidence and the claim	2 points Applies a psychological perspective, theory, concept, or research finding to explain how the evidence supports the claim
	Decision Rules and Scoring Notes		
	Responses that earn 0 points: - The response does not accurately interpret the evidence. - The response identifies evidence without explanation of how it relates to the claim. - The response provides opinions with no reference to the sources provided.	Responses that earn 1 point: The response correctly interprets the evidence in supporting the claim, but does not apply a psychological perspective, theory, concept, or research finding.	Responses that earn 2 points: The response correctly interprets the evidence and applies a psychological perspective, theory, concept, or research finding correctly.
	Examples that earn 0 points: "Secondary school start time should be 9:00 a.m. 28% of students reported falling asleep in school at least once a week." "Secondary school start time should be 9:00 a.m. because I believe this is the best for teenagers."	Examples that earn 1 point: "This evidence supports the claim of starting school later (9 a.m. instead of 8 a.m.) because some high school students have to take care of their younger siblings after school." "This evidence supports the claim of starting school later (9 a.m. instead of 8 a.m.) because it will allow students to have one more hour of sleep."	Examples that earn 2 points: "This evidence supports the claim of starting school later (9 a.m. instead of 8 a.m.) because it will allow students to have one more hour of sleep and potentially reducing the number of times they fall asleep in school. When students have an adequate amount of sleep their ability to consolidate and retrieve memory is better, thus positively affecting their learning."
	Additional Notes: - The explanation and application points can be earned even if the response did not earn the point in B (i) due to incorrectly cited and/or non-specific evidence. - The psychological perspective, theory, concept, or research finding must be explicitly identified in the <i>AP Psychology Course and Exam Description</i> (2024). - The presence of a citation indicates the sentence or phrase is intended to be evidence. All text prior to the citation will be considered evidence. Text after a citation will be considered reasoning, unless accompanied by another citation.		

Reporting Category	Scoring Criteria	
Part C (i) Evidence (0-1 points)	0 point Does not identify nor correctly cite a different piece of specific evidence from either of the other two provided sources to support the claim	1 point Uses a different piece of correctly cited, specific, and relevant evidence from either of the other two provided sources to support the claim
	Decision Rules and Scoring Notes	
	Responses that earn 0 points: - The response identifies no evidence. - The evidence is not correctly cited. - The evidence is nonspecific. - The evidence is from the same source as the one used in Part B. - The response provides evidence not relevant to the claim.	Responses that earn 1 point: The response provides specific and correctly cited evidence relevant to the claim that is from a different source than the one used in Part B.
	Examples that earn 0 points: <i>"Sources say secondary school start time should be 9:00 a.m."</i> [The response has not provided a citation.] <i>"Secondary school start time should be 9 a.m. because people say taking tests before 9 a.m. is very difficult for teenagers."</i> [Test-taking is not a variable discussed in any of the provided sources] <i>"According to Source 1, students should get an additional hour of sleep to avoid sleep debt."</i> [The response uses the same source as used in Part B.]	Examples that earn 1 point: <i>"According to Source 2, students earn higher GPAs when they started school one hour later."</i>
	Additional Notes: Responses that use a correct citation style other than the methods prescribed by the question can earn this point for citing the source.	

Reporting Category	Scoring Criteria		
Part C (ii) Explanation and Application (0-2 points)	0 points Does not explain the relationship between the evidence and the claim	1 point Explains the relationship between the evidence and the claim	2 points Applies a different psychological perspective, theory, concept, or research finding to explain how the evidence supports the claim. The psychological perspective, theory, concept, or research finding is different from the one used in Part B (ii).
	Decision Rules and Scoring Notes		
	Responses that earn 0 points: - The response does not accurately interpret the evidence. - The response identifies evidence without explanation of how it relates to the claim. - The response provides opinions with no reference to the sources provided.	Responses that earn 1 point: - The response correctly interprets the evidence, but does not apply a psychological perspective, theory, concept, or research finding. - The response applies the same psychological perspective, theory, concept, or research finding that was used in Part B; however, the response explains the relationship between the evidence and the claim that does not rely on the application.	Responses that earn 2 points: The response correctly interprets the evidence and applies a psychological perspective, theory, concept, or research finding that is different from the one used in Part B.
	Examples that earn 0 points: "Students should get more sleep because it is good for them." "The evidence clearly supports my claim." "This evidence supports my claim because getting an additional hour of sleep can improve memory consolidation." [The response uses a concept from Part B.]	Examples that earn 1 point: "This evidence supports my claim because getting an additional hour of sleep can improve academic performance, which is a benefit for students."	Examples that earn 2 points: "This evidence supports my claim because a student's circadian rhythm would fall more naturally in the different start times."
	Additional Notes: - The explanation and application points can be earned even if the response did not earn the point in C (i) due to incorrectly cited and/or non-specific evidence. - The explanation and application points can be earned if the response uses the same source in both Parts B and C, provided that the evidence used in Part B and Part C is different. - The psychological perspective, theory, concept, or research finding must be explicitly identified in the <i>AP Psychology Course and Exam Description</i> (2024) and is different from the perspective, theory, concept, or research finding used in Part B. - The presence of a citation indicates the sentence or phrase is intended to be evidence. All text prior to the citation will be considered evidence. Text after a citation will be considered reasoning, unless accompanied by another citation.		